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Enabled Power:
Early Warning and Risk Assessment Tools to help medically at-risk populations and regulators in Texas manage power outages

Isaac Bernhardt Sinaga
Strake Jesuit College Preparatory, Houston, TX 77036, United States of America, e-mail: ibsinaga28@mail.strakejesuit.org

***Abstract*— This research project proposes Enabled Power, a quantitative methodology to help medically at-risk population in Texas managing power outages. Based on the proposed mathematical framework, we develop a set of quantitative tools to classify and quantify risk and impact of outages. The results show a strong correlation between counties with high risk of 24-48 hour outage events and high population of medically-at-risk individuals. We propose a risk-based ranking system that can help regulators to prioritize allocations of government assistance programs. We also propose approach to compute inflection points for each Texas county during major events that can serve as early warnings to medically vulnerable populations and local resources. The analysis shows that early warning can be given as early as 7- 8 hours before peak outage events occur.**

Keywords— data filtering, time series analysis, curvature change, exponential decay, risk analysis, inflection point, event recognition

1. INTRODUCTION

Nearly 200,000 Texans rely on electricity to power their life-saving medical devices every day. Prolonged outages such as 2021 Winter Storm Uri are particularly life-threatening for these communities as described in great detail in [5]. This research project proposes **Enabled Power**, a mathematical framework to help medically at-risk population in Texas managing power outages. To help medically-at risk populations, organizations such as Texas Council for Developmental Disabilities [4] have been campaigning to provision medically at-risk population with home generators and batteries. Allocations of home generators and batteries require prioritization since funding availability is limited. Additionally, as power outage events unfold, an effective early warning system will be very helpful for caregivers and local resources to take early actions.

In this work we aim to develop **quantitative tools** to:

- Recognize, classify and quantify likelihood of power outage events at all Texas counties
- Perform risk-based analysis to prioritize assistance programs for medically-at-risk populations
- Identify an early indicator of prolonged outages based on major event data.

Our analysis used two publicly accessible datasets of power outages and medical populations. The Power Outage Data Information is obtained from Eagle-I data which stores power outage of all US states from 2015 to 2024. The second dataset is from the U.S. Department of Health and Human Services [2], which contains information on medical beneficiaries across the United States. For the purposes of this analysis, we will only focus on the medical beneficiaries within Texas. the mathematical framework we develop here can also be readily applied to other US states.

2. METHODOLOGY

The Enabled Power methodology is implemented in R [8]. **Given** raw data power outage data from [1] and beneficiary data from [2], we performed the following analysis for each Texas county and at state level.

<u>For EACH COUNTY:</u>

1. SELECT time period of interest, in this work we used 2021-2024 data.
2. FILTER raw data. The Eagle Power Outage Data [1] records outage power every 15 minutes and very noisy. We followed the approach of NREL research paper [3] to calculate the average every 4 hours.

3. Develop an algorithm to RECOGNIZE the start time of an outage T_s, the peak time T_{max}, and end time T_{end}
   a. Create a new time series of the derivative from the filtered data:
   $$\frac{dC(t)}{dt} = C(t) - C(t-1)$$
   b. Set $T_{\max}$ as the point when dCt/dt changes sign from positive to negative. Every T_{max} that we find belongs to one outage event.
   c. Set the number of affected customers at peak $C_{tmax} = C(T_{max})$
   d. Find the starting point T_s. We adopted the formula given in [3] that T_s is the time when $C(T_s) \sim 0.1 * C_{tmax}$
   e. We observed that outage restoration is following <u>an exponential decay</u> in which the end time is usually the time

when the final value has decayed by 4 exponents, in this work we set

$$T_{end} \text{ is t when C(t)} \leq (1/e^4) * Ct_{max}$$

f. Compute outage duration as $\mathrm{D}(outage\ event) = T_{end} - T_s$

4. CLASSIFY outage events based on the values of $\mathrm{D}(t)$: Type 1 (<24 hrs.), Type 2 (24-48 hrs.), Type 3 (48-72 hrs.), Type 4 (72-120 hrs.), Type 5 (> 120 hrs.) and plot the histograms.

5. Given the histograms of each outage event classification, for each county we can compute the *likelihood* of each power outage classification as:

$$Likelihood\ (outage\ type\ type) = \frac{number\ of\ outages\ in\ specific\ type}{total\ outage\ events}$$

6. Compute outage RISK FACTOR for each outage type:

$$Risk\ Factor\ (outage\ type) = Likelihood * Impacted \text{ Customers}$$

For STATE level (TEXAS) prioritization :

7. Rank all counties based on RISK FACTOR

a. Counties where there is a high risk of outage and high population medical beneficiary will need to be prioritized during restoration efforts. Government programs such home generator provisioning programs will be useful for counties prone for outages last more than 24 hours while batteries can help for <24 hr outages

For early warning system:

8. Compute INFLECTION POINTS from number of impacted customers and outage duration.

a. The inflection point in calculus indicates a turning point in a curve and mathematically defined is the point at which the second derivative of a curve becomes zero.

b. To compute the inflection point using a built-in function in R, we will use the curve of $\mathrm{D}(outage\ event)$ as a function of affected customers $C(t)$. Due to the numerical disproportion of affected customers against outage duration, we will take the logarithmic scale of the impacted customers.

3. KEY RESULTS

3.1. Application of Enabled Power Methodology

The results of Enabled Power Methodology implementation from filtering to outage type classifications in Figure 1 for Harris County. The methodology automatically identifies outage events based on starting, peak, and end time, resulting in **automatic identification of both minor and major outages**; this is different from works that focus on major outages only.

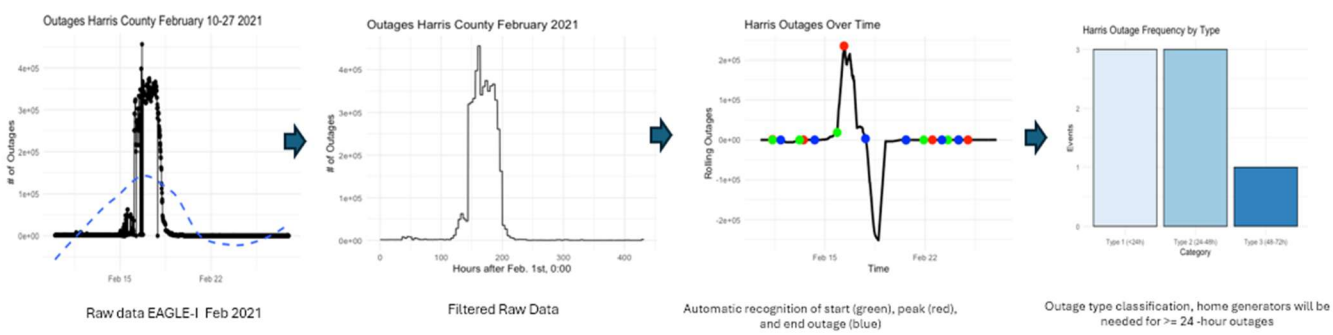

Figure 1 Application of Enabled Power Methodology (Step 1 to 4) to Harris County, Texas

3.2. Correlations between high population of medically vulnerable beneficiaries

The **Type 2 events** (24-48 hours) on the right-hand side of Figure 2 visually correlate quite well the high concentration of medically-at-risk beneficiaries in Texas. This finding shows that to help medically-at-risk communities manage power outages; Texas regulators can prioritize counties based on Type 2 outages, i.e. outages that last between 24-48 hours.

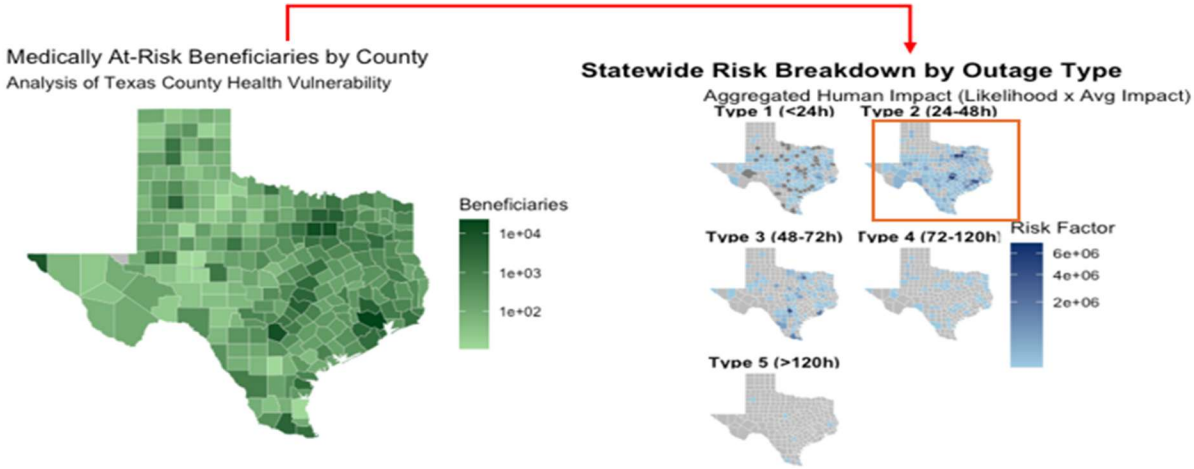

Figure 2 High correlation between Type 2 outage (right) and Texas counties with high beneficiaries (left)

To confirm the visual correlation between Type 2 outages and medically at-risk beneficiary mathematically, we scaled the medical population map into 0 to 1 by dividing each population number by the highest medically at-risk population. The resulting scaled map is shown in Figure 3.

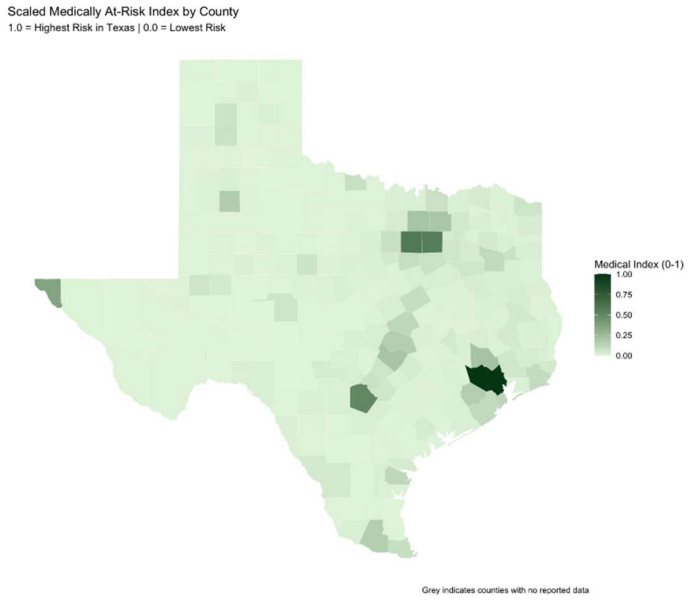

Figure 3 Scaled version of medically at-risk population in Texas

The scaled version of medically at-risk beneficiary is then cross-correlated by the each outage type by multiplying the numerical value in *Figure 3* by the numerical values in each outage type. The cross-correlation results are shown in *Figure 4*, which indeed confirms the visual inspection that Type 2 outage with 24-48 hrs duration is the most common outage that affects medically at-risk populations. That means home generator program is the most useful program for medically-at-risk population. The cross-correlated plot of Type 2 outage shows that several counties are disproportionately affected as indicated by the darkest colors, these are counties that will need to be prioritized for home generator program.

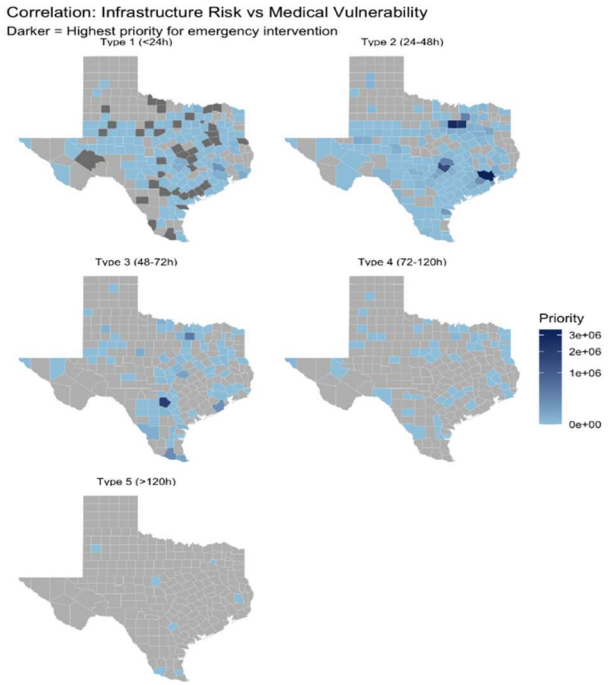

Figure 4 Cross correlation results of medically at-risk population and power outage types. Strongest correlation shown in Type 2 outage (24-48 hrs duration), darkest counties in this region need to be prioritized for home generator provisioning.

3.3. Early warning system based on inflection points

The inflection point computation as proposed in Section 2, Step 8 is shown in Figure 5 for February 2021 period in Harris County. The inflection point corresponds to the time when the outage impacts more than 11, 356 customers, which is in line with DOE classification of "major outage" as the events that affect 10, 000 customers or more. During the Winter Storm Uri period, the inflection points for Harris County occurred about 7 hours prior to the peak outage event, which suggested that we can develop an earlier warning system to alert medically at risk population and local restoration resources 7-8 hours before the peak outage occur.

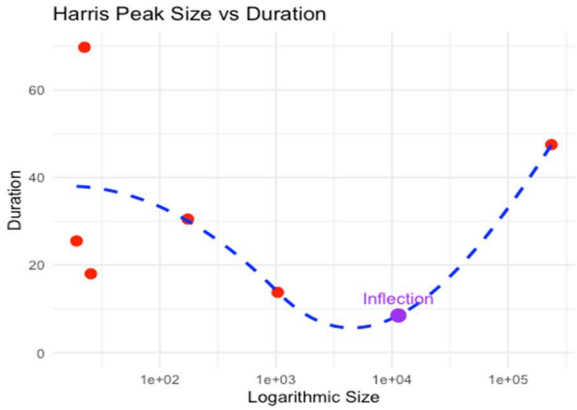

Figure 5 Harris County inflection point calculation (purple point), each red dot represents an outage event

CONCLUSIONS AND FUTURE WORK

We have presented in this paper Enabled Power methodology on large-scale historical power data of Texas. The methodology uses basic calculus and statistical principles that successfully identify minor and major outage events automatically by detecting slope changes, classify outage events and define risk-based analysis of each outage type by combining the likelihood and number of impacted customers for each outage type. We showed that the 24-48 hrs that are affecting medically-at risk population the most. We also showed the premise of the inflection point analysis in calculus to give an early warning of major outages 7-8 hours in advance. This should serve as another early warning for power restoration organizations and family members/caregivers of medically at-risk populations.

For future work, we will improve the robustness of the inflection point analysis by combining multiple major outage events. We will also extend the analysis of inflection points to assess the vulnerability of power system infrastructure in Texas. We also plan to test the Enabled Power methodology to other US states.

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